UNIT 19 ADVANCED TOPICS IN OPTIMISATION– II

Structure

19.0 OBJECTIVES

After going through this unit, you should be able to:

- explain the concept of multiplier;
- examine the envelope theorem in constraint and unconstrained optimisation;
- describe the maxima and minima of a function;
- present the meaning of homogeneous and homothetic functions;
- identify the concave and quasi - concave functions; and
- discuss a few economic applications in advanced optimisation.

19.1 INTRODUCTION

We are aware that there are multiple ways to produce a certain level of output. Similarly, a budget can be spent by a consumer in many ways. But we know that a producer or a consumer prefers the best one out of the many available ways. Such a choice is reached with the aid of constrained optimisation. This unit explains how the Lagrange multiplier technique is a powerful tool for nonlinear optimisation problems by covering both, the equality constrained and inequality constrained nonlinear optimisation problems. Further, the Envelope theorem is discussed given its significance regarding the qualities of differentiability of the value function of a parameterized optimisation problem in economics and the comparative statics of optimisation models. The latter half of the unit presents homothetic and homogenous functions with reference to economic applications and concavity and quasi-concavity with the help of graphical representation and its role in economics.

19.2 MEANING OF MULTIPLIER

This section will provide us an understanding of the equality and inequality constraints in terms of Lagrange's multiplier.

19.2.1 Lagrange's Multiplier Method for Equality Constraints

Without the inequality constraints, the standard form of the nonlinear optimisation problems can be formulated as:

$$\min f(u_1, \cdots, u_n) \qquad \ldots(19.1)$$

Subject to

$$G(u_1, \cdots, u_n) = 0$$

$$H(u_1, \cdots, u_n) \leq 0 \qquad \ldots(19.1a)$$

Where $G = [G_1(u_1, \cdots, u_n) = 0, \cdots, G_k(u_1, \cdots, u_n) = 0]^T$, is the constraints function vector. The Lagrange function F is constructed as

$$F(u, \lambda) = f(u) - \lambda G(u) \qquad \ldots(19.2)$$

where $u = [u_1, \cdots, u_n]$, the variable vector $\lambda = [\lambda_1, \cdots, \lambda_k], \lambda_1, \cdots, \lambda_k$ are called Lagrange multipliers.

The extreme points of the f and the Lagrange multipliers λ satisfy:

$$\nabla F = 0 \qquad \ldots(19.3)$$

That is

$$\frac{\partial F}{\partial u_i} - \sum_{s=1}^{k} \lambda_s \frac{\partial G_s}{\partial u_i} = 0, i = 1, \cdots, n \qquad \ldots(19.4)$$

And $G(u_1, \cdots, u_n) = 0 \qquad \ldots(19.5)$

The prerequisites for constrained nonlinear optimisation problems are defined by the Lagrange multipliers approach.

A) **Mathematical Proof for Lagrange Multiplier Method**

Let us consider the nonlinear optimisation problem which has four variables and two equality constraint

$z = f(u_1, u_2, u_3, u_4)$ …(19.6)

Subject to: $\zeta(u_1, u_2, u_3, u_4) = 0$ …(19.7)

$\varepsilon(u_1, u_2, u_3, u_4) = 0$ …(19.8)

Assuming that at the point $P(\varphi, \phi, \gamma, \tau)$ the function takes an extreme value when compared with the values at all neighboring points satisfying the constraints. Also, assuming that at the point P the Jacobean is

$\frac{\partial(\zeta,\varepsilon)}{\partial(u_3,u_4)} = \zeta_{u_3}\varepsilon_{u4} - \zeta_{u4}\varepsilon_{u_3} \neq 0$ …(19.9)

Thus, at the point of P, we can represent two of the variables, say u_3 and u_4, as functions of the other two, u_1 and u_2, by means (19.7) and (19.8). Substitute the functions $u_3 = g(u_1; u_2)$ and $u_4 = h(u_1; u_2)$ in (19.6), then we get an objective function of the two independent variables u_1 and u_2, and this function must have a free extreme value at the point $u_1 = \varphi$, $u_2 = \phi$ that is its two partial derivatives must be equal to zero at that point. The two equations (19.10) and (19.11) must therefore hold.

$$f_{u_1} + f_{u_3}\frac{\partial u_3}{\partial u_1} + f_t\frac{\partial u_4}{\partial u_1} = 0 \quad \text{…(19.10)}$$

$$f_{u_2} + f_{u_3}\frac{\partial u_3}{\partial u_2} + f_t\frac{\partial u_4}{\partial u_2} = 0 \quad \text{…(19.11)}$$

We can determine two numbers λ and μ in such a way that the two equations

$$f_{u_3} - \lambda\zeta_{u_3} - \mu\varepsilon_{u_3} = 0 \quad \text{…(19.12)}$$

$$f_{u_4} - \lambda\zeta_{u_4} - \mu\varepsilon_{ui_4} = 0 \quad \text{…(19.13)}$$

are satisfied at the point where the extreme value occurs. Take the partial derivative with respect to u_1 on (19.7) and get

$$\zeta_{u_1} + \zeta_{u_3}\frac{\partial u_3}{\partial u_1} + \zeta_{u4}\frac{\partial u_4}{\partial u_1} = 0 \quad \text{…(19.14)}$$

$$\varepsilon_{u_1} + \varepsilon_{u_3}\frac{\partial u_3}{\partial u_1} + \varepsilon_{u4}\frac{\partial u_4}{\partial u_1} = 0 \quad \text{…(19.15)}$$

Multiply (19.14) and (19.15) by $-\lambda$ and $-\mu$ respectively and add them to (19.14), and we get

$$f_{u_1} - \lambda\zeta_{u_1} - \mu\varepsilon_{u_1} + (f_{u_3} - \lambda\zeta_{u_3} - \mu\varepsilon_{u_3})\frac{\partial u_3}{\partial u_1} + (f_{u_4} - \lambda\zeta_{u4} - \mu\varepsilon_{u_4})\frac{\partial u_4}{\partial u_1} = 0$$

…(19.16)

Thus, by definition of λ and μ, we can get

$$f_{u_1} - \lambda\zeta_{u_1} - \mu\varepsilon_{u_1} = 0 \quad \ldots(19.17)$$

Similarly, we get

$$f_{u_2} - \lambda\zeta_{u_2} - \mu\varepsilon_{u2} = 0$$

As a result, we obtain the same conclusion as illustrated in either (19.3) or (19.4) and (19.5). We should apply a similar approach to objective functions that have additional constrains and variables.

19.2.2 Extension to Inequality Constraints

The Lagrange multiplier method also covers the case of inequality constraints, as (19.1(a)). In the feasible region, $G(u_1, \cdots, u_n) = 0$ or $H(u_1, \cdots, u_n) < 0$ when $H_i = 0$. Note that H is said to be active, otherwise H_i is inactive. The augmented Lagrange function is formulated as:

$$F(u, \lambda, \mu) = f(u) - \lambda G(u) - \mu H(u) \quad \ldots(19.18)$$

where, $\mu = [\mu_1, \cdots, \mu_n]$,

$$H = [H_1(u_1, \cdots, u_n), \cdots, H_m(u_1, \cdots, u_n)]^T$$

When H_i is inactive, we can simply remove the constraint by setting $\mu_i = 0$. If $\nabla f = 0$. it points to the descending direction of f. When H_i is active, this direction points out of the feasible region and towards the forbidden side, which means $\nabla H_i > 0$. This is not the solution direction. We can enforce $\mu_i \le 0$ to keep the seeking direction still in the feasible region.

When extended to cover the inequality constraints, the rule for the Lagrange multipliers method can be generalized as:

$$\nabla f(u) - \sum_{i=1}^{k} \lambda_i \nabla G_i(u) - \sum_{j=1}^{m} \mu_j \nabla H_j(u) = 0 \quad \ldots(19.19)$$

$$\mu_i \le 0 \; i = 1, \cdots, m \quad \ldots(19.20)$$

$$\mu_i H_i = 0 \; i = 1, \cdots, m \quad \ldots(19.21)$$

$$G(u) = 0$$

In summary, for inequality constraints, we add them to the Lagrange function just as if they were equality constraints, except that we require that $\mu_i \le 0$ and when $H_i \ne 0$, $\mu_i = 0$. This situation can be compactly expressed as (19.21).

Check Your Progress 1

1) Suppose $u_3 = f(u_1, u_2) = u_1 u_2$ and the constraint is $u_1 + u_2 = 100$. Find maximum value of u_3.

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2) Maximise $u_3 = f(u_1, u_2) = 4u_1^2 + 3u_1 u_2 + 6u_2^2$

subject to $u_1 + u_2 = 56$

Find the values of u_1 and u_2 that maximise u_3. Also find the value of λ.

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19.3 ENVELOPE THEOREM

In economic optimisation problems, the objective functions that we try to maximise/minimise often depend on parameters, like prices. We want to find out how the optimal value is affected by changes in the parameters.

Let us consider the following optimisation problems:

$$\max_u f((u), a)$$

s.t. $g_1(u, a) = 0$

$g_2(u, a) = 0$

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$g_m(u, a) = 0$

Suppose, $r(j) = m.$

Then,

$$L = f(u, a) + \sum_{j=1}^{m} \lambda_j g_j(u, a);$$

where L is the Lagrange function.

If $f^*(a) = \max_x f(u, a)$ is the value function when $u^* = u^*(a)$ is the solution vector, then

$$\frac{\partial f^*(a)}{\partial a_t} = \frac{\partial L(u^*(a),a)}{\partial a_t}, \forall\ t = 1,2,....,k \qquad ...(19.22)$$

Assuming that $f^*(a)$ is differentiable, (19.22) shows that the impact of a change of any parameter or the value function can be found by differentiating the Lagrange function partially with respect the same parameter. This is known as the **Envelope theorem**.

Proof: Given $f^*(a) = f(u^*(a),a)$

$$\frac{\partial f^*(a)}{\partial a_s} = \sum_{i=1}^{n} \frac{\partial f(u^*(a),a)}{\partial u_i} \cdot \frac{\partial u_i(a)}{\partial a_s} + \frac{\partial f(u^*(a),a)}{\partial a_s} \qquad ...(19.23)$$

The first term on the RHS of (19.23) denotes the 2nd order (indirect effect) followed by the 1st order effect (2nd term)

From the FOCs, we have

$$\frac{\partial f(u^*(a),a)}{\partial u_i} = \sum_{j=1}^{m} \lambda_j \frac{\partial g_j(u^*(a),a)}{\partial u_i}\ \forall i \qquad ...(19.24)$$

Using (19.24) in (19.23),

$$\frac{\partial f^*(a)}{\partial a_s} = \sum_{i=1}^{n} \left\{ \sum_{j=1}^{m} \lambda_j \frac{\partial g_j(u^*(a),a)}{\partial u_i} \right\} \frac{\partial u_i^*(a)}{\partial a_s} + \frac{\partial f(u^*(a),a)}{\partial a_s} \qquad ...(19.25)$$

Differentiating the constraint function $g_j(u^*(a),a) = 0$ with respect to a_s, we get

$$\sum_{i=1}^{n} \frac{\partial g_j(u^*(a),a)}{\partial u_i} \frac{\partial u_i^*(a)}{\partial a_s} + \frac{\partial g_j(u^*(a),a)}{\partial a_s} = 0 \qquad ...(19.26)$$

From (19.25), we get

$$\frac{\partial f^*(a)}{\partial a_s} = \sum_{j=1}^{m} \lambda_j \left\{ \sum_{i=1}^{n} \frac{\partial g_j(u^*(a),a)}{\partial u_i} \cdot \frac{\partial u_i^*(a)}{\partial a_s} \right\} + \frac{\partial f^*(u^*(a),a)}{\partial a_s}$$

$$= \sum_{j=1}^{m} \lambda_j \left(-\frac{\partial g_j(u^*(a),a)}{\partial a_s} \right) + \frac{\partial f^*(u^*(a),a)}{\partial a_s} \text{ [from (19.26)]}$$

$$= -\sum_{j=1}^{m} \lambda_j \frac{\partial g_j(u^*(a),a)}{\partial a_s} + \frac{\partial f^*(u^*(a),a)}{\partial a_s}$$

$$= \frac{\partial L(u^*(a),a)}{\partial a_s} \text{ Hence proved.}$$

19.3.1 Envelope Theorem for Unconstrained Optimisation

Suppose that we solve the following optimisation problem:

$\max f(u,v,a)$ where, u is a choice variable and a is a parameter. The first order condition of this problem is

$f(u^*,v^*,a) = 0$.

The optimal choice of u is a function of $a : u^*(a)$. Then the maximum value of f is given by:

$f(u^*(a), v^*(a), a)$, which is only a function of a.

Now, we want to see the effect that a change in a has on $f(u^*(a), v^*(a), a)$.

The Value function for this problem is derived by substituting $(u^*(a), v^*(a))$ into the objective function:

$V(a) = f(u^*(a), v^*(a), a)$, Differentiate w.r.t a we get,

$$\frac{dV}{da} = \frac{\partial f}{\partial u}\frac{\partial u^*}{\partial a} + \frac{\partial f}{\partial v}\frac{dv^*}{da} + \frac{\partial V}{\partial a}.$$

Notice, though that the first order condition of the maximisation states that $\frac{\partial f}{\partial u} = 0 = \frac{\partial f}{\partial v} = 0$ and hence the first two terms in the equation above will vanish, and we are left with

$$\frac{dV}{da} = \frac{\partial V}{\partial a}$$

The partial derivative is evaluated at the point $(u^*(a), v^*(a))$. This result which is called the Envelope Theorem says in words: "The total derivative of the value function with respect to the parameter a is the same as the partial derivative of the objective function evaluated at the optimal point".

Example 19.1: Calculate the value function

Consider the unconstrained problem,

$$\max_{(u_1, u_2)} 4u_1 + au_2 - u_1^2 - u_2^2 + u_1 u_2$$

The FOCs are the following,

$$4 - 2u_1 + u_2 = 0$$
$$a - 2u_2 + u_1 = 0$$

Solving the FOCs, we get,

$$u_1^* = \frac{8+a}{3}, u_2^* = \frac{2a+4}{3}$$

We should check the second order conditions hold so that this candidate solution is a global maximum.

The Value function is

$$V(a) = 4u_1^* + au_2^* - (u_1^*)^2 - (u_2^*)^2 + u_1^* u_2^*$$

$$= 4\frac{8+a}{3} + a\frac{2a+4}{3} - \left[\frac{8+a}{3}\right]^2 - \left[\frac{2a+4}{3}\right]^2 + \frac{(8+a)(2a+4)}{9}$$

19.3.2 Envelope Theorem for Constrained Optimisation

Consider the problem,

$\max f(u,v,\alpha)$

Subject to $g(u,v,\alpha)=0$

The Lagrangian for this problem is:

$L(u,v,\gamma)=f(u,v)+\gamma g(u,v,\alpha)$

Suppose that $(u^*(\alpha),v^*(\alpha),\gamma^*(\alpha))$ solves the constrained optimisation problem. The Value function for this problem is defined as

$V(\alpha)=f(u^*(\alpha),v^*(\alpha),\alpha)$

Let us write the Value function as,

$V(\alpha)=f(u^*(\alpha),v^*(\alpha),\alpha)+\gamma^*(\alpha)g(u^*(\alpha),v^*(\alpha),\alpha)$

Differentiating with respect to α gives,

$$\frac{dV(\alpha)}{d\alpha}=\frac{\partial f}{\partial u}\frac{du^*}{d\alpha}+\frac{\partial f}{\partial v}\frac{dv^*}{d\alpha}+\frac{\partial f}{\partial \alpha}+g(u^*(\alpha),v^*(\alpha),\alpha)\frac{d\gamma^*}{d\alpha}$$

$$+\gamma^*(\alpha)\left[\frac{\partial g}{\partial u}\frac{du^*}{d\alpha}+\frac{\partial g}{\partial v}\frac{dv^*}{d\alpha}+\frac{\partial g}{\partial \alpha}\right]$$

This can be written as,

$$\frac{dV(\alpha)}{d\alpha}=\left[\frac{\partial f}{\partial u}+\gamma^*\frac{\partial g}{\partial u}\right]\frac{du^*}{d\alpha}+\left[\frac{\partial f}{\partial v}+\gamma^*\frac{\partial g}{\partial v}\right]\frac{dv^*}{d\alpha}+g(u^*(\alpha),v^*(\alpha),\alpha)\frac{d\gamma^*}{d\alpha}+\frac{\partial f}{\partial \alpha}+\gamma^*\frac{\partial g}{\partial \alpha}$$

. Note that the first two terms on the right-hand side drop out because (u^*,v^*,γ^*) must satisfy the necessary conditions for constrained optimisation. The third term drops out because $g(u^*(\alpha),v^*(\alpha),\alpha)=0$

We are left with the following:

$$\frac{dV(\alpha)}{d\alpha}=\frac{\partial f}{\partial \alpha}+\gamma^*\frac{\partial g}{\partial \alpha}=\frac{\partial L}{\partial \alpha}(u^*,v^*,\gamma^*)$$

The above equation shows that the derivative of the value function with respect to the parameter α is the partial derivative of the Lagrangian function with respect to α evaluated at (u^*,v^*,γ^*)

The envelope theorem is important in the field of restricted optimisation since it allows us to compute the partial derivative of the Lagrangian with respect to the parameter. We can use the theorem to determine how the maximum value function changes when one or more parameters are changed.

Check Your Progress 2

1) Maximise the function $f(u;a) = -u^2 + 2au + 4a^2$ with respect to $a > 0$.

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2) Consider the following Lagrange problem:

Maximise $f(u,v) = u + 3v$

subject to $g(u,v) = u^2 + av^2 = 10.$

Use the envelope theorem to estimate the maximum value $f^*(1.01)$ when $a = 1.01$, and check this by computing the optimal value function $f^*(a)$.

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19.4 HOMOGENEOUS AND HOMOTHETIC FUNCTIONS

In this section we will understand about the homogenous function and homothetic functions with the help of definitions, examples and theorems.

19.4.1 Homogeneous Functions and Properties

Homogeneous functions are widely present in economics. Profit functions and cost functions that are derived from production functions, as well as the demand functions which are derived from utility functions, are homogeneous in the standard economic theory.

Definition: A real valued function $f(u_1, u_2, ..., u_n)$ is homogeneous of degree k if for all $t > 0$

$f(tu_1, tu_2, ..., tu_n) = t^k f(u_1, u_2, ..., u_n)$

Economic Applications

A. Returns to Scale

Constant return to scale - production function which is homogeneous of degree $k = 1$.

Increasing return to scale - production function which is homogeneous of degree $k > 1$.

Decreasing return to scale - production function which is homogeneous of degree $k < 1$.

B. Cobb-Douglas utility function: $u(x_1, x_2, \ldots, x_n) = A x_1^{\alpha_1} \ldots\ldots x_n^{\alpha_n}$ is homogeneous of degree $k = \alpha_1 + \ldots + \alpha_n$.

C. Constant elasticity of substitution (CES) function: $U(a_1 x_1^p + a_2 x_2^p)^{\frac{q}{p}}$ is homogeneous of degree q.

Demand functions that are derived from utility functions are homogeneous of degree 0: if the prices $(p_1, p_2, \ldots, p_n)$ and income I change say 10 times then the demand will not change. More precisely, let $U(x, x_2, \ldots, x_n)$ be the utility function, $p = (p_1, p_2, \ldots, p_n)$ be the price vector, $x = (x_1, x_2, \ldots, x_n)$ be a consumption bundle and let $p \bullet x = p_1 x_1 + p_2 x + \ldots + p_n x_n \le I$ be the budget constraint. The demand function $x = D(p_1, p_2, \ldots, p_n, I)$ associates to each price vector p and income level I the consumption bundle x which is a solution of the following constrained maximisation problem

Maximise $U(x)$

subject to $p_1 x_1 + p_2 x_2 + \ldots + p_n x_n \le I$.

So, we claim that demand function $x = D(p_1, p_2, \ldots, p_n, I)$ is homogeneous of degree 0, that is

$D(tp_1, \ldots, tp_n, tI) = t^0 D(p_1, \ldots, p_n, I) = D(p_1, \ldots, p_n, I)$.

The bundle $D(tp_1, \ldots, tp_n, tI)$ is a solution of the following constrained maximisation problem

Maximise$U(x)$

subject to $tp_1 x_1 + tp_2 x_2 + \ldots + tp_n x_n \le tI$

As we see this is the same problem thus,

$D(tp_1, \ldots, tp_n, tI) = t^0 D(p_1, \ldots, p_n, I) = D(p_1, \ldots, p_n, I)$

Properties of Homogeneous Functions

Theorem A: If $f(u_1, u_2, \ldots, u_n)$ is homogeneous of degree k then its first order partial derivatives are homogeneous of degree $k-1$.

Proof: Differentiate $f(tu_1, tu_2, \ldots, tu_n) = t^k f(u_1, u_2, \ldots, u_n)$ by u_i

Theorem B: Level sets of a homogeneous function are radial expansions of one another, that is for arbitrary $u = (u_1, u_2, \ldots, u_n)$, $v = (v_1, v_2, \ldots, v_n)$, $t > 0$ if u and v are on the same level set then their radial expansions tu and tv are on the same level set too.

Proof: We must show that $f(u) = f(v)$ implies $f(tu) = f(tv)$. Indeed $f(tu) = t^k f(tu) = t^k f(v) = f(tv)$.

Theorem C: Suppose a function of $n+1$ variables $F(u_1, u_2, \ldots, u_n, z)$ is homogeneous of degree k, that is

$F(tu_1, tu_2, \ldots, tu_n, tz) = t^k F(u_1, u_2, \ldots, u_n, z)$ and let $f(u_1, u_2, \ldots, u_n)$ be the function of n variables which is the restriction of F on arguments with $z = 1$